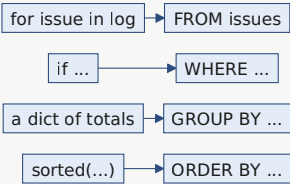


1 Each Python move has a clause



The librarian's three questions, how many issues, which department borrows most and how many days late, came out the same in Python and in SQL.

Crux. A loop over records becomes `FROM`, its `if` becomes `WHERE`, and its dictionary of totals becomes `GROUP BY`.

2 Written in one order, run in another

You write	Postgres runs
SELECT	FROM
FROM	WHERE
WHERE	GROUP BY
GROUP BY	HAVING
HAVING	SELECT
ORDER BY	ORDER BY

A clause out of written order stops with `ERROR: syntax error at or near`, naming the first word Postgres could not place.

Crux. Write in the fixed order, and reason in the run order.

3 Which rows

Condition	Keeps
dept = 'Arts'	text in single quotes
days_kept > 14 AND week = 1	both true
dept = 'Arts' OR dept = 'Maths'	either true
book IN ('Poetry', 'History')	one of a list
days_kept BETWEEN 10 AND 14	a range, both ends included

Double quotes name a column, so `dept = "Arts"` stops with `ERROR: column "Arts" does not exist`.

Crux. Single quotes for text, double quotes never for text.

4 One row per group, then groups that pass

```
SELECT dept, COUNT(*) AS issues, SUM(days_kept) AS days_out
FROM issues
WHERE days_kept > 14           -- rows, before the groups
GROUP BY dept                 -- one row per department
HAVING COUNT(*) >= 2          -- groups, after they exist
ORDER BY issues DESC, dept;   -- a second column settles ties
```

`COUNT(*)`, `SUM`, `MIN`, `MAX` and `AVG` each turn many rows into one number.

Crux. `WHERE` filters rows, `GROUP BY` makes the groups, and `HAVING` filters the groups.

5 Postgres in your codespace

```
sudo service postgresql start # after every restart
psql -d library               # open the database
psql -d library -f setup.sql  # run a whole file
```

Inside `psql`, end every query with `;`, leave with `\q`, and press `q` to close a long result.

`connection to server on socket ... failed` means the server is not running: start it and try again.

Crux. Start the server, open the database, and end every query with a semicolon.